

mobile robot is a self-contained automaton that can wholly move with respect to its environment. There is no direct way to measure a mobile robot's position instantaneously. Instead, one must integrate the motion of the robot over time. Add to this the inaccuracies of *motion estimation* due to slippage and it is clear that measuring a mobile robot's position precisely is an extremely challenging task.

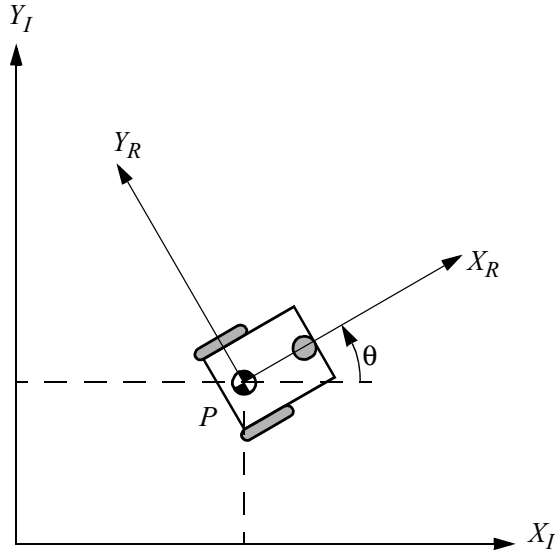
The process of understanding the motions of a robot begins with the process of describing the contribution each wheel provides for motion. Each wheel has a role in enabling the whole robot to move. By the same token, each wheel also imposes constraints on the robot's motion; for example, refusing to skid laterally. In the following section, we introduce notation that allows expression of robot motion in a global reference frame as well as the robot's local reference frame. Then, using this notation, we demonstrate the construction of simple forward kinematic models of motion, describing how the robot as a whole moves as a function of its geometry and individual wheel behavior. Next, we formally describe the kinematic constraints of individual wheels, and then combine these kinematic constraints to express the whole robot's kinematic constraints. With these tools, one can evaluate the paths and trajectories that define the robot's maneuverability.

## 3.2 Kinematic Models and Constraints

Deriving a model for the whole robot's motion is a bottom-up process. Each individual wheel contributes to the robot's motion and, at the same time, imposes constraints on robot motion. Wheels are tied together based on robot chassis geometry, and therefore their constraints combine to form constraints on the overall motion of the robot chassis. But the forces and constraints of each wheel must be expressed with respect to a clear and consistent reference frame. This is particularly important in mobile robotics because of its self-contained and mobile nature; a clear mapping between global and local frames of reference is required. We begin by defining these reference frames formally, then using the resulting formalism to annotate the kinematics of individual wheels and whole robots. Throughout this process we draw extensively on the notation and terminology presented in [52].

### 3.2.1 Representing robot position

Throughout this analysis we model the robot as a rigid body on wheels, operating on a horizontal plane. The total dimensionality of this robot chassis on the plane is three, two for position in the plane and one for orientation along the vertical axis, which is orthogonal to the plane. Of course, there are additional degrees of freedom and flexibility due to the wheel axles, wheel steering joints, and wheel castor joints. However by robot *chassis* we refer only to the rigid body of the robot, ignoring the joints and degrees of freedom internal to the robot and its wheels.

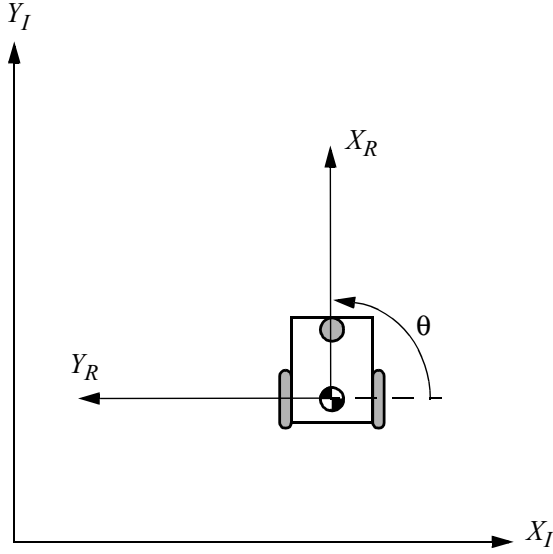
**Figure 3.1**

The global reference frame and the robot local reference frame.

In order to specify the position of the robot on the plane we establish a relationship between the global reference frame of the plane and the local reference frame of the robot, as in figure 3.1. The axes  $X_I$  and  $Y_I$  define an arbitrary inertial basis on the plane as the global reference frame from some origin  $O$ :  $\{X_I, Y_I\}$ . To specify the position of the robot, choose a point  $P$  on the robot chassis as its position reference point. The basis  $\{X_R, Y_R\}$  defines two axes relative to  $P$  on the robot chassis and is thus the robot's local reference frame. The position of  $P$  in the global reference frame is specified by coordinates  $x$  and  $y$ , and the angular difference between the global and local reference frames is given by  $\theta$ . We can describe the pose of the robot as a vector with these three elements. Note the use of the subscript  $I$  to clarify the basis of this pose as the global reference frame:

$$\xi_I = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} \quad (3.1)$$

To describe robot motion in terms of component motions, it will be necessary to map motion along the axes of the global reference frame to motion along the axes of the robot's local reference frame. Of course, the mapping is a function of the current pose of the robot. This mapping is accomplished using the *orthogonal rotation matrix*:

**Figure 3.2**

The mobile robot aligned with a global axis.

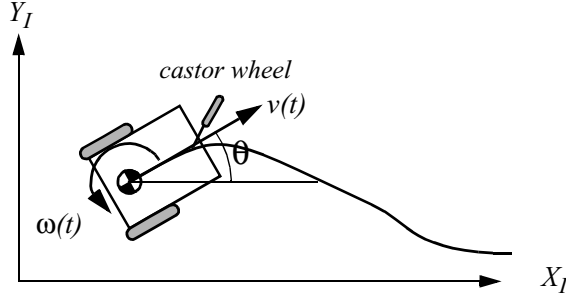
$$R(\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3.2)$$

This matrix can be used to map motion in the global reference frame  $\{X_I, Y_I\}$  to motion in terms of the local reference frame  $\{X_R, Y_R\}$ . This operation is denoted by  $R(\theta)\dot{\xi}_I$  because the computation of this operation depends on the value of  $\theta$ :

$$\dot{\xi}_R = R\left(\frac{\pi}{2}\right)\dot{\xi}_I \quad (3.3)$$

For example, consider the robot in figure 3.2. For this robot, because  $\theta = \frac{\pi}{2}$  we can easily compute the instantaneous rotation matrix  $R$ :

$$R\left(\frac{\pi}{2}\right) = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3.4)$$

**Figure 3.3**

A differential-drive robot in its global reference frame.

Given some velocity  $(\dot{x}, \dot{y}, \dot{\theta})$  in the global reference frame we can compute the components of motion along this robot's local axes  $X_R$  and  $Y_R$ . In this case, due to the specific angle of the robot, motion along  $X_R$  is equal to  $\dot{y}$  and motion along  $Y_R$  is  $-\dot{x}$ :

$$\dot{\xi}_R = R\left(\frac{\pi}{2}\right)\dot{\xi}_I = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{y} \\ -\dot{x} \\ \dot{\theta} \end{bmatrix} \quad (3.5)$$

### 3.2.2 Forward kinematic models

In the simplest cases, the mapping described by equation (3.3) is sufficient to generate a formula that captures the forward kinematics of the mobile robot: how does the robot move, given its geometry and the speeds of its wheels? More formally, consider the example shown in figure 3.3.

This differential drive robot has two wheels, each with diameter  $r$ . Given a point  $P$  centered between the two drive wheels, each wheel is a distance  $l$  from  $P$ . Given  $r$ ,  $l$ ,  $\theta$ , and the spinning speed of each wheel,  $\dot{\phi}_1$  and  $\dot{\phi}_2$ , a forward kinematic model would predict the robot's overall speed in the global reference frame:

$$\dot{\xi}_I = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = f(l, r, \theta, \dot{\phi}_1, \dot{\phi}_2) \quad (3.6)$$

From equation (3.3) we know that we can compute the robot's motion in the global reference frame from motion in its local reference frame:  $\dot{\xi}_I = R(\theta)^{-1}\dot{\xi}_R$ . Therefore, the strategy will be to first compute the contribution of each of the two wheels in the local reference

frame,  $\dot{\xi}_R$ . For this example of a differential-drive chassis, this problem is particularly straightforward.

Suppose that the robot's local reference frame is aligned such that the robot moves forward along  $+X_R$ , as shown in figure 3.1. First consider the contribution of each wheel's spinning speed to the translation speed at  $P$  in the direction of  $+X_R$ . If one wheel spins while the other wheel contributes nothing and is stationary, since  $P$  is halfway between the two wheels, it will move instantaneously with half the speed:  $\dot{x}_{r1} = (1/2)r\dot{\phi}_1$  and  $\dot{x}_{r2} = (1/2)r\dot{\phi}_2$ . In a differential drive robot, these two contributions can simply be added to calculate the  $x_R$  component of  $\dot{\xi}_R$ . Consider, for example, a differential robot in which each wheel spins with equal speed but in opposite directions. The result is a stationary, spinning robot. As expected,  $\dot{x}_R$  will be zero in this case. The value of  $\dot{y}_R$  is even simpler to calculate. Neither wheel can contribute to sideways motion in the robot's reference frame, and so  $\dot{y}_R$  is always zero. Finally, we must compute the rotational component  $\dot{\theta}_R$  of  $\dot{\xi}_R$ . Once again, the contributions of each wheel can be computed independently and just added. Consider the right wheel (we will call this wheel 1). Forward spin of this wheel results in *counterclockwise* rotation at point  $P$ . Recall that if wheel 1 spins alone, the robot pivots around wheel 2. The rotation velocity  $\omega_1$  at  $P$  can be computed because the wheel is instantaneously moving along the arc of a circle of radius  $2l$ :

$$\omega_1 = \frac{r\dot{\phi}_1}{2l} \quad (3.7)$$

The same calculation applies to the left wheel, with the exception that forward spin results in *clockwise* rotation at point  $P$ :

$$\omega_2 = \frac{-r\dot{\phi}_2}{2l} \quad (3.8)$$

Combining these individual formulas yields a kinematic model for the differential-drive example robot:

$$\dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} \frac{r\dot{\phi}_1}{2} + \frac{r\dot{\phi}_2}{2} \\ 0 \\ \frac{r\dot{\phi}_1}{2l} + \frac{-r\dot{\phi}_2}{2l} \end{bmatrix} \quad (3.9)$$

We can now use this kinematic model in an example. However, we must first compute  $R(\theta)^{-1}$ . In general, calculating the inverse of a matrix may be challenging. In this case, however, it is easy because it is simply a transform from  $\xi_R$  to  $\xi_I$  rather than vice versa:

$$R(\theta)^{-1} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3.10)$$

Suppose that the robot is positioned such that  $\theta = \pi/2$ ,  $r = 1$ , and  $l = 1$ . If the robot engages its wheels unevenly, with speeds  $\dot{\phi}_1 = 4$  and  $\dot{\phi}_2 = 2$ , we can compute its velocity in the global reference frame:

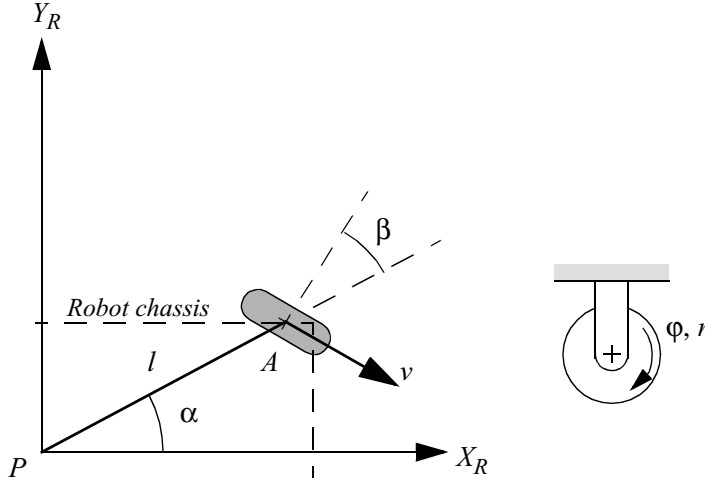
$$\dot{\xi}_I = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} \quad (3.11)$$

So this robot will move instantaneously along the  $y$ -axis of the global reference frame with speed 3 while rotating with speed 1. This approach to kinematic modeling can provide information about the motion of a robot given its component wheel speeds in straightforward cases. However, we wish to determine the space of possible motions for each robot chassis design. To do this, we must go further, describing formally the constraints on robot motion imposed by each wheel. Section 3.2.3 begins this process by describing constraints for various wheel types; the rest of this chapter provides tools for analyzing the characteristics and workspace of a robot given these constraints.

### 3.2.3 Wheel kinematic constraints

The first step to a kinematic model of the robot is to express constraints on the motions of individual wheels. Just as shown in section 3.2.2, the motions of individual wheels can later be combined to compute the motion of the robot as a whole. As discussed in chapter 2, there are four basic wheel types with widely varying kinematic properties. Therefore, we begin by presenting sets of constraints specific to each wheel type.

However, several important assumptions will simplify this presentation. We assume that the plane of the wheel always remains vertical and that there is in all cases one single point of contact between the wheel and the ground plane. Furthermore, we assume that there is no sliding at this single point of contact. That is, the wheel undergoes motion only under conditions of pure rolling and rotation about the vertical axis through the contact point. For a more thorough treatment of kinematics, including sliding contact, refer to [25].

**Figure 3.4**

A fixed standard wheel and its parameters.

Under these assumptions, we present two constraints for every wheel type. The first constraint enforces the concept of rolling contact – that the wheel must roll when motion takes place in the appropriate direction. The second constraint enforces the concept of no lateral slippage – that the wheel must not slide orthogonal to the wheel plane.

### 3.2.3.1 Fixed standard wheel

The fixed standard wheel has no vertical axis of rotation for steering. Its angle to the chassis is thus fixed, and it is limited to motion back and forth along the wheel plane and rotation around its contact point with the ground plane. Figure 3.4 depicts a fixed standard wheel  $A$  and indicates its position pose relative to the robot's local reference frame  $\{X_R, Y_R\}$ . The position of  $A$  is expressed in polar coordinates by distance  $l$  and angle  $\alpha$ . The angle of the wheel plane relative to the chassis is denoted by  $\beta$ , which is fixed since the fixed standard wheel is not steerable. The wheel, which has radius  $r$ , can spin over time, and so its rotational position around its horizontal axle is a function of time  $t$ :  $\varphi(t)$ .

The rolling constraint for this wheel enforces that all motion along the direction of the wheel plane must be accompanied by the appropriate amount of wheel spin so that there is pure rolling at the contact point:

$$\left[ \sin(\alpha + \beta) - \cos(\alpha + \beta) (-l) \cos \beta \right] R(\theta) \dot{\xi}_l - r \dot{\varphi} = 0 \quad (3.12)$$

The first term of the sum denotes the total motion along the wheel plane. The three elements of the vector on the left represent mappings from each of  $\dot{x}, \dot{y}, \dot{\theta}$  to their contributions for motion along the wheel plane. Note that the  $R(\theta)\dot{\xi}_I$  term is used to transform the motion parameters  $\dot{\xi}_I$  that are in the global reference frame  $\{X_I, Y_I\}$  into motion parameters in the local reference frame  $\{X_R, Y_R\}$  as shown in example equation (3.5). This is necessary because all other parameters in the equation,  $\alpha, \beta, l$ , are in terms of the robot's local reference frame. This motion along the wheel plane must be equal, according to this constraint, to the motion accomplished by spinning the wheel,  $r\dot{\phi}$ .

The sliding constraint for this wheel enforces that the component of the wheel's motion orthogonal to the wheel plane must be zero:

$$\begin{bmatrix} \cos(\alpha + \beta) & \sin(\alpha + \beta) & l \sin \beta \end{bmatrix} R(\theta) \dot{\xi}_I = 0 \quad (3.13)$$

For example, suppose that wheel  $A$  is in a position such that  $\{(\alpha = 0), (\beta = 0)\}$ . This would place the contact point of the wheel on  $X_I$  with the plane of the wheel oriented parallel to  $Y_I$ . If  $\theta = 0$ , then the sliding constraint [equation (3.13)] reduces to

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = 0 \quad (3.14)$$

This constrains the component of motion along  $X_I$  to be zero and since  $X_I$  and  $X_R$  are parallel in this example, the wheel is constrained from sliding sideways, as expected.

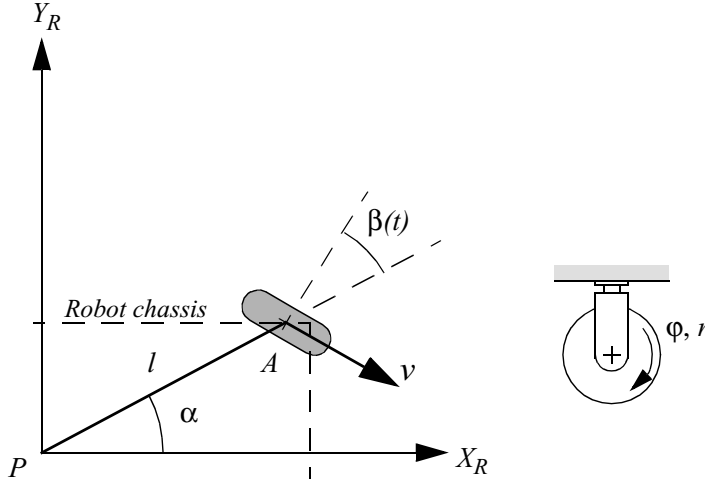
### 3.2.3.2 Steered standard wheel

The steered standard wheel differs from the fixed standard wheel only in that there is an additional degree of freedom: the wheel may rotate around a vertical axis passing through the center of the wheel and the ground contact point. The equations of position for the steered standard wheel (figure 3.5) are identical to that of the fixed standard wheel shown in figure 3.4 with one exception. The orientation of the wheel to the robot chassis is no longer a single fixed value,  $\beta$ , but instead varies as a function of time:  $\beta(t)$ . The rolling and sliding constraints are

$$\begin{bmatrix} \sin(\alpha + \beta) & -\cos(\alpha + \beta) & (-l) \cos \beta \end{bmatrix} R(\theta) \dot{\xi}_I - r\dot{\phi} = 0 \quad (3.15)$$

$$\begin{bmatrix} \cos(\alpha + \beta) & \sin(\alpha + \beta) & l \sin \beta \end{bmatrix} R(\theta) \dot{\xi}_I = 0 \quad (3.16)$$





**Figure 3.5**  
A steered standard wheel and its parameters.

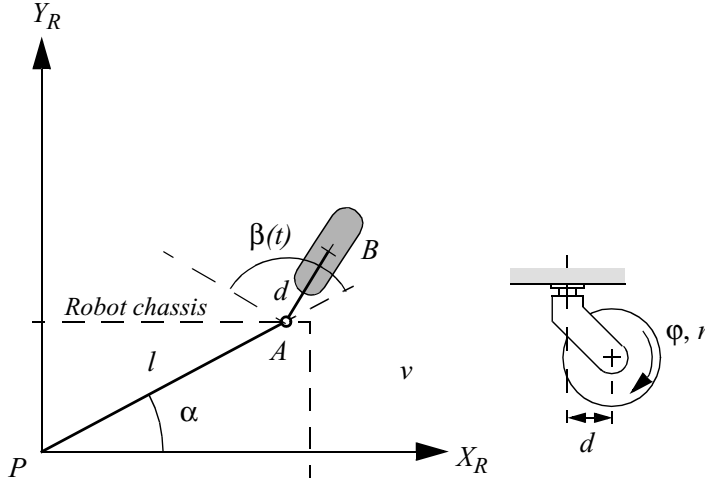
These constraints are identical to those of the fixed standard wheel because, unlike  $\dot{\phi}$ ,  $\dot{\beta}$  does not have a direct impact on the instantaneous motion constraints of a robot. It is only by integrating over time that changes in steering angle can affect the mobility of a vehicle. This may seem subtle, but is a very important distinction between change in steering position,  $\dot{\beta}$ , and change in wheel spin,  $\dot{\phi}$ .

### 3.2.3.3 Castor wheel

Castor wheels are able to steer around a vertical axis. However, unlike the steered standard wheel, the vertical axis of rotation in a castor wheel does not pass through the ground contact point. Figure 3.6 depicts a castor wheel, demonstrating that formal specification of the castor wheel's position requires an additional parameter.

The wheel contact point is now at position  $B$ , which is connected by a rigid rod  $AB$  of fixed length  $d$  to point  $A$  fixes the location of the vertical axis about which  $B$  steers, and this point  $A$  has a position specified in the robot's reference frame, as in figure 3.6. We assume that the plane of the wheel is aligned with  $AB$  at all times. Similar to the steered standard wheel, the castor wheel has two parameters that vary as a function of time.  $\phi(t)$  represents the wheel spin over time as before.  $\beta(t)$  denotes the steering angle and orientation of  $AB$  over time.

For the castor wheel, the rolling constraint is identical to equation (3.15) because the offset axis plays no role during motion that is aligned with the wheel plane:



**Figure 3.6**  
A castor wheel and its parameters.

$$\begin{bmatrix} \sin(\alpha + \beta) & -\cos(\alpha + \beta) & (-l) \cos \beta \end{bmatrix} R(\theta) \dot{\xi}_I - r \dot{\phi} = 0 \quad (3.17)$$

The castor geometry does, however, have significant impact on the sliding constraint. The critical issue is that the lateral force on the wheel occurs at point  $A$  because this is the attachment point of the wheel to the chassis. Because of the offset ground contact point relative to  $A$ , the constraint that there be zero lateral movement would be wrong. Instead, the constraint is much like a rolling constraint, in that appropriate rotation of the vertical axis must take place:

$$\begin{bmatrix} \cos(\alpha + \beta) & \sin(\alpha + \beta) & d + l \sin \beta \end{bmatrix} R(\theta) \dot{\xi}_I + d \dot{\beta} = 0 \quad (3.18)$$

In equation (3.18), any motion orthogonal to the wheel plane must be balanced by an equivalent and opposite amount of castor steering motion. This result is critical to the success of castor wheels because by setting the value of  $\dot{\beta}$  any arbitrary lateral motion can be acceptable. In a steered standard wheel, the steering action does not by itself cause a movement of the robot chassis. But in a castor wheel the steering action itself moves the robot chassis because of the offset between the ground contact point and the vertical axis of rotation.



**Figure 3.7**  
Office chair with five castor wheels.

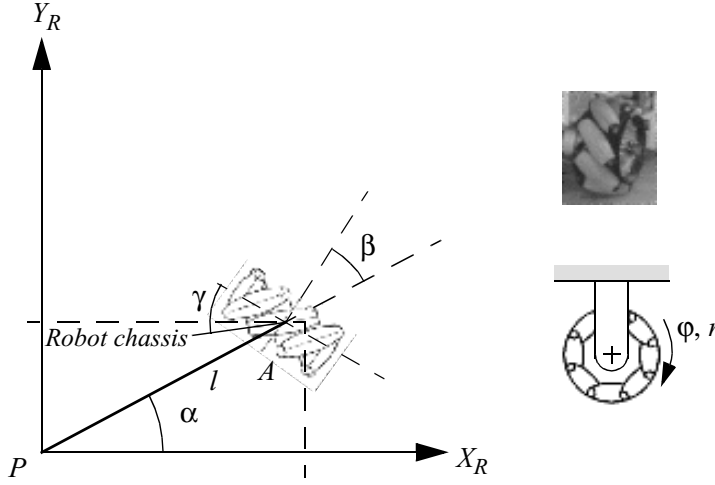
More concisely, it can be surmised from equations (3.17) and (3.18) that, given *any* robot chassis motion  $\xi_I$ , there exists some value for spin speed  $\dot{\phi}$  and steering speed  $\dot{\beta}$  such that the constraints are met. Therefore, a robot with only castor wheels can move with any velocity in the space of possible robot motions. We term such systems *omnidirectional*.

A real-world example of such a system is the five-castor wheel office chair shown in figure 3.7. Assuming that all joints are able to move freely, you may select any motion vector on the plane for the chair and push it by hand. Its castor wheels will spin and steer as needed to achieve that motion without contact point sliding. By the same token, if each of the chair's castor wheels housed two motors, one for spinning and one for steering, then a control system would be able to move the chair along any trajectory in the plane. Thus, although the kinematics of castor wheels is somewhat complex, such wheels do not impose any real constraints on the kinematics of a robot chassis.

#### 3.2.3.4 Swedish wheel

Swedish wheels have no vertical axis of rotation, yet are able to move *omnidirectionally* like the castor wheel. This is possible by adding a degree of freedom to the fixed standard wheel. Swedish wheels consist of a fixed standard wheel with rollers attached to the wheel perimeter with axes that are antiparallel to the main axis of the fixed wheel component. The exact angle  $\gamma$  between the roller axes and the main axis can vary, as shown in figure 3.8.

For example, given a Swedish 45-degree wheel, the motion vectors of the principal axis and the roller axes can be drawn as in figure 3.8. Since each axis can spin clockwise or counterclockwise, one can combine any vector along one axis with any vector along the other axis. These two axes are not necessarily independent (except in the case of the Swedish 90-degree wheel); however, it is visually clear that any desired direction of motion is achievable by choosing the appropriate two vectors.



**Figure 3.8**  
A Swedish wheel and its parameters.

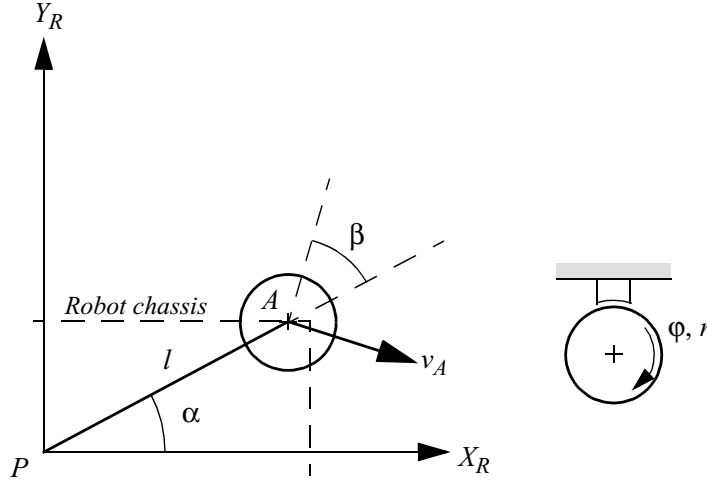
The pose of a Swedish wheel is expressed exactly as in a fixed standard wheel, with the addition of a term,  $\gamma$ , representing the angle between the main wheel plane and the axis of rotation of the small circumferential rollers. This is depicted in figure 3.8 within the robot's reference frame.

Formulating the constraint for a Swedish wheel requires some subtlety. The instantaneous constraint is due to the specific orientation of the small rollers. The axis around which these rollers spin is a zero component of velocity at the contact point. That is, moving in that direction without spinning the main axis is not possible without sliding. The motion constraint that is derived looks identical to the rolling constraint for the fixed standard wheel in equation (3.12) except that the formula is modified by adding  $\gamma$  such that the effective direction along which the rolling constraint holds is along this zero component rather than along the wheel plane:

$$\left[ \sin(\alpha + \beta + \gamma) \quad -\cos(\alpha + \beta + \gamma) \quad (-l) \cos(\beta + \gamma) \right] R(\theta) \dot{\xi}_I - r \dot{\phi} \cos \gamma = 0 \quad (3.19)$$

Orthogonal to this direction the motion is not constrained because of the free rotation  $\dot{\phi}_{sw}$  of the small rollers.

$$\left[ \cos(\alpha + \beta + \gamma) \quad \sin(\alpha + \beta + \gamma) \quad l \sin(\beta + \gamma) \right] R(\theta) \dot{\xi}_I - r \dot{\phi} \sin \gamma - r_{sw} \dot{\phi}_{sw} = 0 \quad (3.20)$$

**Figure 3.9**

A spherical wheel and its parameters.

The behavior of this constraint and thereby the Swedish wheel changes dramatically as the value  $\gamma$  varies. Consider  $\gamma = 0$ . This represents the swedish 90-degree wheel. In this case, the zero component of velocity is in line with the wheel plane and so equation (3.19) reduces exactly to equation (3.12), the fixed standard wheel rolling constraint. But because of the rollers, there is no sliding constraint orthogonal to the wheel plane [see equation (3.20)]. By varying the value of  $\phi$ , any desired motion vector can be made to satisfy equation (3.19) and therefore the wheel is omnidirectional. In fact, this special case of the Swedish design results in fully decoupled motion, in that the rollers and the main wheel provide orthogonal directions of motion.

At the other extreme, consider  $\gamma = \pi/2$ . In this case, the rollers have axes of rotation that are parallel to the main wheel axis of rotation. Interestingly, if this value is substituted for  $\gamma$  in equation (3.19) the result is the fixed standard wheel sliding constraint, equation (3.13). In other words, the rollers provide no benefit in terms of lateral freedom of motion since they are simply aligned with the main wheel. However, in this case the main wheel never needs to spin and therefore the rolling constraint disappears. This is a degenerate form of the Swedish wheel and therefore we assume in the remainder of this chapter that  $\gamma \neq \pi/2$ .

### 3.2.3.5 Spherical wheel

The final wheel type, a ball or spherical wheel, places no direct constraints on motion (figure 3.9). Such a mechanism has no principal axis of rotation, and therefore no appropriate rolling or sliding constraints exist. As with castor wheels and Swedish wheels, the spherical

wheel is clearly omnidirectional and places no constraints on the robot chassis kinematics. Therefore equation (3.21) simply describes the roll rate of the ball in the direction of motion  $v_A$  of point  $A$  of the robot.

$$\left[ \sin(\alpha + \beta) - \cos(\alpha + \beta) (-l) \cos \beta \right] R(\theta) \dot{\xi}_I - r \dot{\phi} = 0 \quad (3.21)$$

By definition the wheel rotation orthogonal to this direction is zero.

$$\left[ \cos(\alpha + \beta) \sin(\alpha + \beta) l \sin \beta \right] R(\theta) \dot{\xi}_I = 0 \quad (3.22)$$

As can be seen, the equations for the spherical wheel are exactly the same as for the fixed standard wheel. However, the interpretation of equation (3.22) is different. The omnidirectional spherical wheel can have any arbitrary direction of movement, where the motion direction given by  $\beta$  is a free variable deduced from equation (3.22). Consider the case that the robot is in pure translation in the direction of  $Y_R$ . Then equation (3.22) reduces to  $\sin(\alpha + \beta) = 0$ , thus  $\beta = -\alpha$ , which makes sense for this special case.

### 3.2.4 Robot kinematic constraints

Given a mobile robot with  $M$  wheels we can now compute the kinematic constraints of the robot chassis. The key idea is that each wheel imposes zero or more constraints on robot motion, and so the process is simply one of appropriately combining all of the kinematic constraints arising from all of the wheels based on the placement of those wheels on the robot chassis.

We have categorized all wheels into five categories: (1) fixed and (2) steerable standard wheels, (3) castor wheels, (4) Swedish wheels, and (5) spherical wheels. But note from the wheel kinematic constraints in equations (3.17), (3.18), and (3.19) that the castor wheel, Swedish wheel, and spherical wheel impose *no* kinematic constraints on the robot chassis, since  $\dot{\xi}_I$  can range freely in all of these cases owing to the internal wheel degrees of freedom.

Therefore only fixed standard wheels and steerable standard wheels have impact on robot chassis kinematics and therefore require consideration when computing the robot's kinematic constraints. Suppose that the robot has a total of  $N$  standard wheels, comprising  $N_f$  fixed standard wheels and  $N_s$  steerable standard wheels. We use  $\beta_s(t)$  to denote the variable steering angles of the  $N_s$  steerable standard wheels. In contrast,  $\beta_f$  refers to the orientation of the  $N_f$  fixed standard wheels as depicted in figure 3.4. In the case of wheel spin, both the fixed and steerable wheels have rotational positions around the horizontal axle that vary as a function of time. We denote the fixed and steerable cases separately as  $\varphi_f(t)$  and  $\varphi_s(t)$ , and use  $\varphi(t)$  as an aggregate matrix that combines both values:

$$\varphi(t) = \begin{bmatrix} \varphi_f(t) \\ \varphi_s(t) \end{bmatrix} \quad (3.23)$$

The rolling constraints of all wheels can now be collected in a single expression:

$$J_1(\beta_s)R(\theta)\dot{\xi}_I - J_2\dot{\phi} = 0 \quad (3.24)$$

This expression bears a strong resemblance to the rolling constraint of a single wheel, but substitutes matrices in lieu of single values, thus taking into account all wheels.  $J_2$  is a constant diagonal  $N \times N$  matrix whose entries are radii  $r$  of all standard wheels.  $J_1(\beta_s)$  denotes a matrix with projections for all wheels to their motions along their individual wheel planes:

$$J_1(\beta_s) = \begin{bmatrix} J_{1f} \\ J_{1s}(\beta_s) \end{bmatrix} \quad (3.25)$$

Note that  $J_1(\beta_s)$  is only a function of  $\beta_s$  and not  $\beta_f$ . This is because the orientations of steerable standard wheels vary as a function of time, whereas the orientations of fixed standard wheels are constant.  $J_{1f}$  is therefore a constant matrix of projections for all fixed standard wheels. It has size  $(N_f \times 3)$ , with each row consisting of the three terms in the three-matrix from equation (3.12) for each fixed standard wheel.  $J_{1s}(\beta_s)$  is a matrix of size  $(N_s \times 3)$ , with each row consisting of the three terms in the three-matrix from equation (3.15) for each steerable standard wheel.

In summary, equation (3.24) represents the constraint that all standard wheels must spin around their horizontal axis an appropriate amount based on their motions along the wheel plane so that rolling occurs at the ground contact point.

We use the same technique to collect the sliding constraints of all standard wheels into a single expression with the same structure as equations (3.13) and (3.16):

$$C_1(\beta_s)R(\theta)\dot{\xi}_I = 0 \quad (3.26)$$

$$C_1(\beta_s) = \begin{bmatrix} C_{1f} \\ C_{1s}(\beta_s) \end{bmatrix} \quad (3.27)$$

$C_{1f}$  and  $C_{1s}$  are  $(N_f \times 3)$  and  $(N_s \times 3)$  matrices whose rows are the three terms in the three-matrix of equations (3.13) and (3.16) for all fixed and steerable standard wheels. Thus

equation (3.26) is a constraint over all standard wheels that their components of motion orthogonal to their wheel planes must be zero. This sliding constraint over all standard wheels has the most significant impact on defining the overall maneuverability of the robot chassis, as explained in the next section.

### 3.2.5 Examples: robot kinematic models and constraints

In section 3.2.2 we presented a forward kinematic solution for  $\dot{\xi}_I$  in the case of a simple differential-drive robot by combining each wheel's contribution to robot motion. We can now use the tools presented above to construct the same kinematic expression by direct application of the rolling constraints for every wheel type. We proceed with this technique applied again to the differential drive robot, enabling verification of the method as compared to the results of section 3.2.2. Then we proceed to the case of the three-wheeled omnidirectional robot.

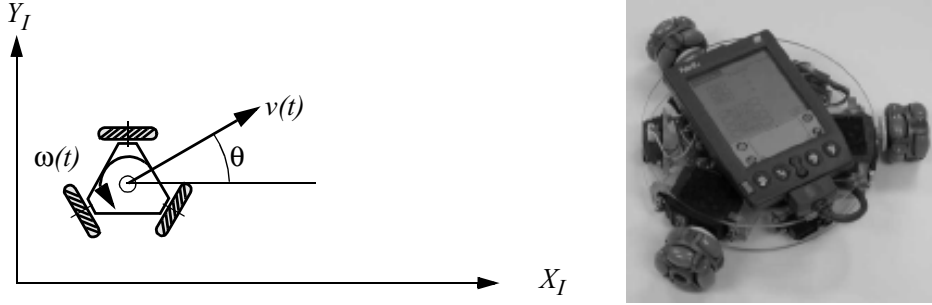
#### 3.2.5.1 A differential-drive robot example

First, refer to equations (3.24) and (3.26). These equations relate robot motion to the rolling and sliding constraints  $J_1(\beta_s)$  and  $C_1(\beta_s)$ , and the wheel spin speed of the robot's wheels,  $\dot{\phi}$ . Fusing these two equations yields the following expression:

$$\begin{bmatrix} J_1(\beta_s) \\ C_1(\beta_s) \end{bmatrix} R(\theta) \dot{\xi}_I = \begin{bmatrix} J_2 \dot{\phi} \\ 0 \end{bmatrix} \quad (3.28)$$

Once again, consider the differential drive robot in figure 3.3. We will construct  $J_1(\beta_s)$  and  $C_1(\beta_s)$  directly from the rolling constraints of each wheel. The castor is unpowered and is free to move in any direction, so we ignore this third point of contact altogether. The two remaining drive wheels are not steerable, and therefore  $J_1(\beta_s)$  and  $C_1(\beta_s)$  simplify to  $J_{1f}$  and  $C_{1f}$  respectively. To employ the fixed standard wheel's rolling constraint formula, equation (3.12), we must first identify each wheel's values for  $\alpha$  and  $\beta$ . Suppose that the robot's local reference frame is aligned such that the robot moves forward along  $+X_R$ , as shown in figure 3.1. In this case, for the right wheel  $\alpha = -\pi/2$ ,  $\beta = \pi$ , and for the left wheel,  $\alpha = \pi/2$ ,  $\beta = 0$ . Note the value of  $\beta$  for the right wheel is necessary to ensure that positive spin causes motion in the  $+X_R$  direction (figure 3.4). Now we can compute the  $J_{1f}$  and  $C_{1f}$  matrix using the matrix terms from equations (3.12) and (3.13). Because the two fixed standard wheels are parallel, equation (3.13) results in only one independent equation, and equation (3.28) gives



**Figure 3.10**

A three-wheel omnidrive robot developed by Carnegie Mellon University ([www.cs.cmu.edu/~pprk](http://www.cs.cmu.edu/~pprk)).

$$\begin{bmatrix} 1 & 0 & l \\ 1 & 0 & -l \\ 0 & 1 & 0 \end{bmatrix} R(\theta) \dot{\xi}_I = \begin{bmatrix} J_2 \phi \\ 0 \end{bmatrix} \quad (3.29)$$

Inverting equation (3.29) yields the kinematic equation specific to our differential drive robot:

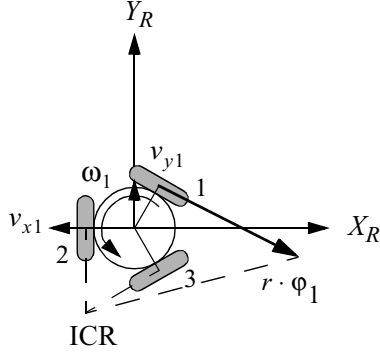
$$\dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} 1 & 0 & l \\ 1 & 0 & -l \\ 0 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} J_2 \phi \\ 0 \end{bmatrix} = R(\theta)^{-1} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \\ \frac{1}{2l} & -\frac{1}{2l} & 0 \end{bmatrix} \begin{bmatrix} J_2 \phi \\ 0 \end{bmatrix} \quad (3.30)$$

This demonstrates that, for the simple differential-drive case, the combination of wheel rolling and sliding constraints describes the kinematic behavior, based on our manual calculation in section 3.2.2.

### 3.2.5.2 An omnidirectional robot example

Consider the omniwheel robot shown in figure 3.10. This robot has three Swedish 90-degree wheels, arranged radially symmetrically, with the rollers perpendicular to each main wheel.

First we must impose a specific local reference frame upon the robot. We do so by choosing point  $P$  at the center of the robot, then aligning the robot with the local reference

**Figure 3.11**

The local reference frame plus detailed parameters for wheel 1.

frame such that  $X_R$  is colinear with the axis of wheel 2. Figure 3.11 shows the robot and its local reference frame arranged in this manner.

We assume that the distance between each wheel and  $P$  is  $l$ , and that all three wheels have the same radius,  $r$ . Once again, the value of  $\dot{\xi}_I$  can be computed as a combination of the rolling constraints of the robot's three omnidirectional wheels, as in equation (3.28). As with the differential-drive robot, since this robot has no steerable wheels,  $J_1(\beta_s)$  simplifies to  $J_{1f}$ :

$$\dot{\xi}_I = R(\theta)^{-1} J_{1f}^{-1} J_2 \dot{\phi} \quad (3.31)$$

We calculate  $J_{1f}$  using the matrix elements of the rolling constraints for the Swedish wheel, given by equation (3.19). But to use these values, we must establish the values  $\alpha$ ,  $\beta$ ,  $\gamma$  for each wheel. Referring to figure (3.8), we can see that  $\gamma = 0$  for the Swedish 90-degree wheel. Note that this immediately simplifies equation (3.19) to equation (3.12), the rolling constraints of a fixed standard wheel. Given our particular placement of the local reference frame, the value of  $\alpha$  for each wheel is easily computed:  $(\alpha_1 = \pi/3)$ ,  $(\alpha_2 = \pi)$ ,  $(\alpha_3 = -\pi/3)$ . Furthermore,  $\beta = 0$  for all wheels because the wheels are tangent to the robot's circular body. Constructing and simplifying  $J_{1f}$  using equation (3.12) yields

$$J_{1f} = \begin{bmatrix} \sin \frac{\pi}{3} & -\cos \frac{\pi}{3} & -l \\ 0 & -\cos \pi & -l \\ \sin -\frac{\pi}{3} & -\cos -\frac{\pi}{3} & -l \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & -l \\ 0 & 1 & -l \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & -l \end{bmatrix} \quad (3.32)$$

Once again, computing the value of  $\dot{\xi}_I$  requires calculating the inverse,  $J_{1f}^{-1}$ , as needed in equation (3.31). One approach would be to apply rote methods for calculating the inverse of a 3 x 3 square matrix. A second approach would be to compute the contribution of each Swedish wheel to chassis motion, as shown in section 3.2.2. We leave this process as an exercise for the enthusiast. Once the inverse is obtained,  $\dot{\xi}_I$  can be isolated:

$$\dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{3}} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3l} & -\frac{1}{3l} & -\frac{1}{3l} \end{bmatrix} J_2 \dot{\phi} \quad (3.33)$$

Consider a specific omnidrive chassis with  $l = 1$  and  $r = 1$  for all wheels. The robot's local reference frame and global reference frame are aligned, so that  $\theta = 0$ . If wheels 1, 2, and 3 spin at speeds  $(\phi_1 = 4)$ ,  $(\phi_2 = 1)$ ,  $(\phi_3 = 2)$ , what is the resulting motion of the whole robot? Using the equation above, the answer can be calculated readily:

$$\dot{\xi}_I = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{3}} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{3}} \\ -\frac{4}{3} \\ -\frac{7}{3} \end{bmatrix} \quad (3.34)$$

So this robot will move instantaneously along the  $x$ -axis with positive speed and along the  $y$  axis with negative speed while rotating clockwise. We can see from the above examples that robot motion can be predicted by combining the rolling constraints of individual wheels.

The sliding constraints comprising  $C_1(\beta_s)$  can be used to go even further, enabling us to evaluate the maneuverability and workspace of the robot rather than just its predicted motion. Next, we examine methods for using the sliding constraints, sometimes in conjunction with rolling constraints, to generate powerful analyses of the maneuverability of a robot chassis.

### 3.3 Mobile Robot Maneuverability

The kinematic mobility of a robot chassis is its ability to directly move in the environment. The basic constraint limiting mobility is the rule that every wheel must satisfy its sliding constraint. Therefore, we can formally derive robot mobility by starting from equation (3.26).

In addition to instantaneous kinematic motion, a mobile robot is able to further manipulate its position, over time, by steering steerable wheels. As we will see in section 3.3.3, the overall maneuverability of a robot is thus a combination of the mobility available based on the kinematic sliding constraints of the standard wheels, plus the additional freedom contributed by steering and spinning the steerable standard wheels.

#### 3.3.1 Degree of mobility

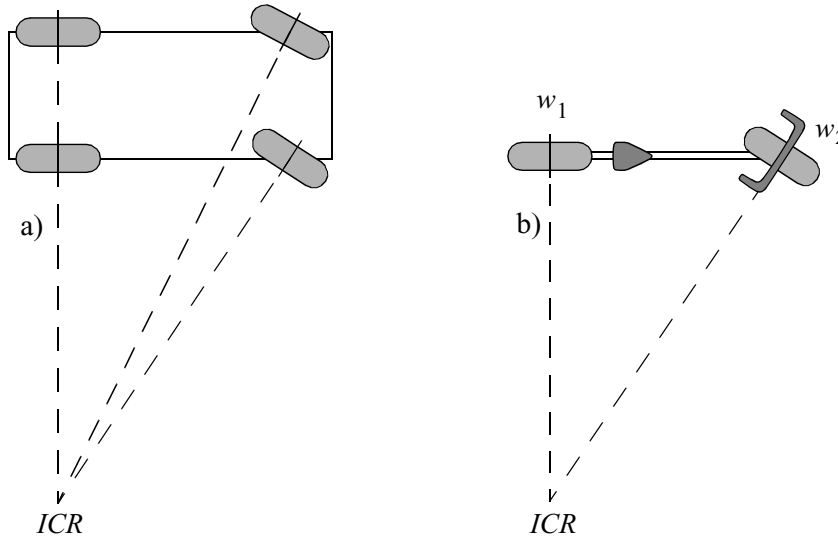
Equation (3.26) imposes the constraint that every wheel must avoid any lateral slip. Of course, this holds separately for each and every wheel, and so it is possible to specify this constraint separately for fixed and for steerable standard wheels:

$$C_{1f}R(\theta)\dot{\xi}_I = 0 \quad (3.35)$$

$$C_{1s}(\beta_s)R(\theta)\dot{\xi}_I = 0 \quad (3.36)$$

For both of these constraints to be satisfied, the motion vector  $R(\theta)\dot{\xi}_I$  must belong to the *null space* of the projection matrix  $C_1(\beta_s)$ , which is simply a combination of  $C_{1f}$  and  $C_{1s}$ . Mathematically, the null space of  $C_1(\beta_s)$  is the space  $N$  such that for any vector  $n$  in  $N$ ,  $C_1(\beta_s)n = 0$ . If the kinematic constraints are to be honored, then the motion of the robot must always be within this space  $N$ . The kinematic constraints [equations (3.35) and (3.36)] can also be demonstrated geometrically using the concept of a robot's *instantaneous center of rotation (ICR)*.

Consider a single standard wheel. It is forced by the sliding constraint to have zero lateral motion. This can be shown geometrically by drawing a *zero motion line* through its horizontal axis, perpendicular to the wheel plane (figure 3.12). At any given instant, wheel motion along the zero motion line must be zero. In other words, the wheel must be moving

**Figure 3.12**

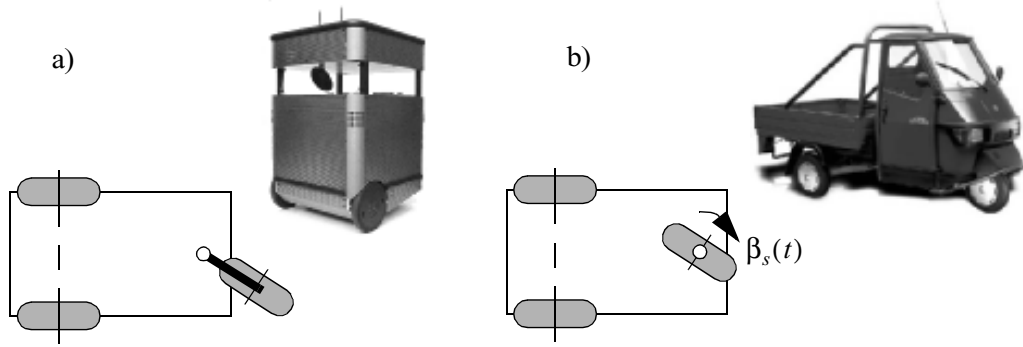
(a) Four-wheel with car-like Ackerman steering. (b) bicycle.

instantaneously along some circle of radius  $R$  such that the center of that circle is located on the zero motion line. This center point, called the instantaneous center of rotation, may lie anywhere along the zero motion line. When  $R$  is at infinity, the wheel moves in a straight line.

A robot such as the Ackerman vehicle in figure 3.12a can have several wheels, but must always have a single  $ICR$ . Because all of its zero motion lines meet at a single point, there is a single solution for robot motion, placing the  $ICR$  at this meet point.

This  $ICR$  geometric construction demonstrates how robot mobility is a function of the number of constraints on the robot's motion, not the number of wheels. In figure 3.12b, the bicycle shown has two wheels,  $w_1$  and  $w_2$ . Each wheel contributes a constraint, or a zero motion line. Taken together the two constraints result in a single point as the only remaining solution for the  $ICR$ . This is because the two constraints are independent, and thus each further constrains overall robot motion.

But in the case of the differential drive robot in figure 3.13a, the two wheels are aligned along the same horizontal axis. Therefore, the  $ICR$  is constrained to lie along a line, not at a specific point. In fact, the second wheel imposes no additional kinematic constraints on robot motion since its zero motion line is identical to that of the first wheel. Thus, although the bicycle and differential-drive chassis have the same number of nonomnidirectional wheels, the former has two independent kinematic constraints while the latter has only one.

**Figure 3.13**

(a) Differential drive robot with two individually motorized wheels and a castor wheel, e.g., the Pygmalion robot at EPFL. (b) Tricycle with two fixed standard wheels and one steered standard wheel, e.g. Piaggio minitransporter.

The Ackerman vehicle of figure 3.12a demonstrates another way in which a wheel may be unable to contribute an independent constraint to the robot kinematics. This vehicle has two steerable standard wheels. Given the instantaneous position of just one of these steerable wheels and the position of the fixed rear wheels, there is only a single solution for the  $ICR$ . The position of the second steerable wheel is absolutely constrained by the  $ICR$ . Therefore, it offers no independent constraints to robot motion.

Robot chassis kinematics is therefore a function of the set of *independent* constraints arising from all standard wheels. The mathematical interpretation of independence is related to the *rank* of a matrix. Recall that the rank of a matrix is the smallest number of independent rows or columns. Equation (3.26) represents all sliding constraints imposed by the wheels of the mobile robot. Therefore  $rank[C_1(\beta_s)]$  is the number of independent constraints.

The greater the number of independent constraints, and therefore the greater the rank of  $C_1(\beta_s)$ , the more constrained is the mobility of the robot. For example, consider a robot with a single fixed standard wheel. Remember that we consider only standard wheels. This robot may be a unicycle or it may have several Swedish wheels; however, it has exactly one fixed standard wheel. The wheel is at a position specified by parameters  $\alpha, \beta, l$  relative to the robot's local reference frame.  $C_1(\beta_s)$  is comprised of  $C_{1f}$  and  $C_{1s}$ . However, since there are no steerable standard wheels  $C_{1s}$  is empty and therefore  $C_1(\beta_s)$  contains only  $C_{1f}$ . Because there is one fixed standard wheel, this matrix has a rank of one and therefore this robot has a single independent constrain on mobility:

$$C_1(\beta_s) = C_{1f} = \begin{bmatrix} \cos(\alpha + \beta) & \sin(\alpha + \beta) & l \sin \beta \end{bmatrix} \quad (3.37)$$

Now let us add an additional fixed standard wheel to create a differential-drive robot by constraining the second wheel to be aligned with the same horizontal axis as the original wheel. Without loss of generality, we can place point  $P$  at the midpoint between the centers of the two wheels. Given  $\alpha_1, \beta_1, l_1$  for wheel  $w_1$  and  $\alpha_2, \beta_2, l_2$  for wheel  $w_2$ , it holds geometrically that  $\{(l_1 = l_2), (\beta_1 = \beta_2 = 0), (\alpha_1 + \pi = \alpha_2)\}$ . Therefore, in this case, the matrix  $C_1(\beta_s)$  has two constraints but a rank of one:

$$C_1(\beta_s) = C_{1f} = \begin{bmatrix} \cos(\alpha_1) & \sin(\alpha_1) & 0 \\ \cos(\alpha_1 + \pi) & \sin(\alpha_1 + \pi) & 0 \end{bmatrix} \quad (3.38)$$

Alternatively, consider the case when  $w_2$  is placed in the wheel plane of  $w_1$  but with the same orientation, as in a bicycle with the steering locked in the forward position. We again place point  $P$  between the two wheel centers, and orient the wheels such that they lie on axis  $x_1$ . This geometry implies that  $\{(l_1 = l_2), (\beta_1 = \beta_2 = \pi/2), (\alpha_1 = 0), (\alpha_2 = \pi)\}$  and, therefore, the matrix  $C_1(\beta_s)$  retains two independent constraints and has a rank of two:

$$C_1(\beta_s) = C_{1f} = \begin{bmatrix} \cos(\pi/2) & \sin(\pi/2) & l_1 \sin(\pi/2) \\ \cos(3\pi/2) & \sin(3\pi/2) & l_1 \sin(\pi/2) \end{bmatrix} = \begin{bmatrix} 0 & 1 & l_1 \\ 0 & -1 & l_1 \end{bmatrix} \quad (3.39)$$

In general, if  $\text{rank}[C_{1f}] > 1$  then the vehicle can, at best, only travel along a circle or along a straight line. This configuration means that the robot has two or more independent constraints due to fixed standard wheels that do not share the same horizontal axis of rotation. Because such configurations have only a degenerate form of mobility in the plane, we do not consider them in the remainder of this chapter. Note, however, that some degenerate configurations such as the four-wheeled slip/skid steering system are useful in certain environments, such as on loose soil and sand, even though they fail to satisfy sliding constraints. Not surprisingly, the price that must be paid for such violations of the sliding constraints is that dead reckoning based on odometry becomes less accurate and power efficiency is reduced dramatically.

In general, a robot will have zero or more fixed standard wheels and zero or more steerable standard wheels. We can therefore identify the possible range of rank values for any robot:  $0 \leq \text{rank}[C_1(\beta_s)] \leq 3$ . Consider the case  $\text{rank}[C_1(\beta_s)] = 0$ . This is only possible if there are zero independent kinematic constraints in  $C_1(\beta_s)$ . In this case there are neither fixed nor steerable standard wheels attached to the robot frame:  $N_f = N_s = 0$ .

Consider the other extreme,  $\text{rank}[C_1(\beta_s)] = 3$ . This is the maximum possible rank since the kinematic constraints are specified along three degrees of freedom (i.e., the constraint matrix is three columns wide). Therefore, there cannot be more than three indepen-

dent constraints. In fact, when  $\text{rank}[C_1(\beta_s)] = 3$ , then the robot is completely constrained in all directions and is, therefore, degenerate since motion in the plane is totally impossible.

Now we are ready to formally define a robot's *degree of mobility*  $\delta_m$ :

$$\delta_m = \dim N[C_1(\beta_s)] = 3 - \text{rank}[C_1(\beta_s)] \quad (3.40)$$

The dimensionality of the null space ( $\dim N$ ) of matrix  $C_1(\beta_s)$  is a measure of the number of degrees of freedom of the robot chassis that can be immediately manipulated through changes in wheel velocity. It is logical therefore that  $\delta_m$  must range between 0 and 3.

Consider an ordinary differential-drive chassis. On such a robot there are two fixed standard wheels sharing a common horizontal axis. As discussed above, the second wheel adds no independent kinematic constraints to the system. Therefore,  $\text{rank}[C_1(\beta_s)] = 1$  and  $\delta_m = 2$ . This fits with intuition: a differential drive robot can control both the rate of its change in orientation and its forward/reverse speed, *simply by manipulating wheel velocities*. In other words, its *ICR* is constrained to lie on the infinite line extending from its wheels' horizontal axes.

In contrast, consider a bicycle chassis. This configuration consists of one fixed standard wheel and one steerable standard wheel. In this case, each wheel contributes an independent sliding constraint to  $C_1(\beta_s)$ . Therefore,  $\delta_m = 1$ . Note that the bicycle has the same total number of nonomidirectional wheels as the differential-drive chassis, and indeed one of its wheels is steerable. Yet it has one less degree of mobility. Upon reflection this is appropriate. A bicycle only has control over its forward/reverse speed by direct manipulation of wheel velocities. Only by steering can the bicycle change its *ICR*.

As expected, based on equation (3.40) any robot consisting only of omnidirectional wheels such as Swedish or spherical wheels will have the maximum mobility,  $\delta_m = 3$ . Such a robot can directly manipulate all three degrees of freedom.

### 3.3.2 Degree of steerability

The degree of mobility defined above quantifies the degrees of controllable freedom based on changes to wheel velocity. Steering can also have an eventual impact on a robot chassis pose  $\xi$ , although the impact is indirect because after changing the angle of a steerable standard wheel, the robot must move for the change in steering angle to have impact on pose.

As with mobility, we care about the number of independently controllable steering parameters when defining the *degree of steerability*  $\delta_s$ :

$$\delta_s = \text{rank}[C_{1s}(\beta_s)] \quad (3.41)$$



Recall that in the case of mobility, an increase in the rank of  $C_1(\beta_s)$  implied more kinematic constraints and thus a less mobile system. In the case of steerability, an increase in the rank of  $C_{1s}(\beta_s)$  implies more degrees of steering freedom and thus greater eventual maneuverability. Since  $C_1(\beta_s)$  includes  $C_{1s}(\beta_s)$ , this means that a steered standard wheel can both decrease mobility and increase steerability: its particular orientation at any instant imposes a kinematic constraint, but its ability to change that orientation can lead to additional trajectories.

The range of  $\delta_s$  can be specified:  $0 \leq \delta_s \leq 2$ . The case  $\delta_s = 0$  implies that the robot has no steerable standard wheels,  $N_s = 0$ . The case  $\delta_s = 1$  is most common when a robot configuration includes one or more steerable standard wheels.

For example, consider an ordinary automobile. In this case  $N_f = 2$  and  $N_s = 2$ . But the fixed wheels share a common axle and so  $\text{rank}[C_{1f}] = 1$ . The fixed wheels and any one of the steerable wheels constrain the *ICR* to be a point along the line extending from the rear axle. Therefore, the second steerable wheel cannot impose any independent kinematic constraint and so  $\text{rank}[C_{1s}(\beta_s)] = 1$ . In this case  $\delta_m = 1$  and  $\delta_s = 1$ .

The case  $\delta_s = 2$  is only possible in robots with no fixed standard wheels:  $N_f = 0$ . Under these circumstances, it is possible to create a chassis with two separate steerable standard wheels, like a pseudobicycle (or the two-steer) in which both wheels are steerable. Then, orienting one wheel constrains the *ICR* to a line while the second wheel can constrain the *ICR* to any point along that line. Interestingly, this means that the  $\delta_s = 2$  implies that the robot can place its *ICR* anywhere on the ground plane.

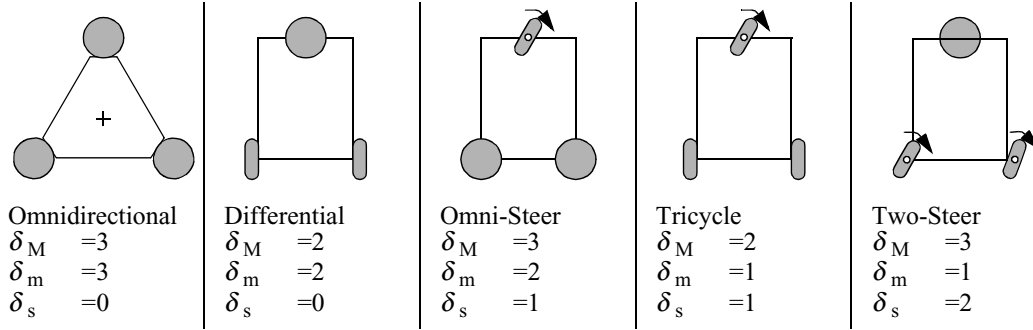
### 3.3.3 Robot maneuverability

The overall degrees of freedom that a robot can manipulate, called the *degree of maneuverability*  $\delta_M$ , can be readily defined in terms of mobility and steerability:

$$\delta_M = \delta_m + \delta_s \quad (3.42)$$

Therefore maneuverability includes both the degrees of freedom that the robot manipulates directly through wheel velocity and the degrees of freedom that it indirectly manipulates by changing the steering configuration and moving. Based on the investigations of the previous sections, one can draw the basic types of wheel configurations. They are depicted in figure 3.14

Note that two robots with the same  $\delta_M$  are not necessarily equivalent. For example, differential drive and tricycle geometries (figure 3.13) have equal maneuverability  $\delta_M = 2$ . In differential drive all maneuverability is the result of direct mobility because  $\delta_m = 2$  and  $\delta_s = 0$ . In the case of a tricycle the maneuverability results from steering also:  $\delta_m = 1$  and  $\delta_s = 1$ . Neither of these configurations allows the *ICR* to range anywhere on the plane. In both cases, the *ICR* must lie on a predefined line with respect to the robot refer-

**Figure 3.14**

The five basic types of three-wheel configurations. The spherical wheels can be replaced by castor or Swedish wheels without influencing maneuverability. More configurations with various numbers of wheels are found in chapter 2.

ence frame. In the case of differential drive, this line extends from the common axle of the two fixed standard wheels, with the differential wheel velocities setting the *ICR* point on this line. In a tricycle, this line extends from the shared common axle of the fixed wheels, with the steerable wheel setting the *ICR* point along this line.

More generally, for any robot with  $\delta_M = 2$  the *ICR* is always constrained to lie on a line and for any robot with  $\delta_M = 3$  the *ICR* can be set to any point on the plane.

One final example will demonstrate the use of the tools we have developed above. One common robot configuration for indoor mobile robotics research is the *synchro drive* configuration (figure 2.22). Such a robot has two motors and three wheels that are locked together. One motor provides power for spinning all three wheels while the second motor provides power for steering all three wheels. In a three-wheeled synchro drive robot  $N_f = 0$  and  $N_s = 3$ . Therefore  $\text{rank}[C_{1s}(\beta_s)]$  can be used to determine both  $\delta_m$  and  $\delta_s$ . The three wheels do not share a common axle, therefore two of the three contribute independent sliding constraints. The third must be dependent on these two constraints for motion to be possible. Therefore  $\text{rank}[C_{1s}(\beta_s)] = 2$  and  $\delta_m = 1$ . This is intuitively correct. A synchro drive robot with the steering frozen manipulates only one degree of freedom, consisting of traveling back and forth on a straight line.

However an interesting complication occurs when considering  $\delta_s$ . Based on equation (3.41) the robot should have  $\delta_s = 2$ . Indeed, for a three-wheel-steering robot with the geometric configuration of a synchro drive robot this would be correct. However, we have additional information: in a synchro drive configuration a single motor steers all three wheels using a belt drive. Therefore, although ideally, if the wheels were independently steerable, then the system would achieve  $\delta_s = 2$ , in the case of synchro drive the drive

system further constrains the kinematics such that in reality  $\delta_s = 1$ . Finally, we can compute maneuverability based on these values:  $\delta_M = 2$  for a synchro drive robot.

This result implies that a synchro drive robot can only manipulate, in total, two degrees of freedom. In fact, if the reader reflects on the wheel configuration of a synchro drive robot it will become apparent that there is no way for the chassis orientation to change. Only the  $x - y$  position of the chassis can be manipulated and so, indeed, a synchro drive robot has only two degrees of freedom, in agreement with our mathematical conclusion.

### 3.4 Mobile Robot Workspace

For a robot, maneuverability is equivalent to its control degrees of freedom. But the robot is situated in some environment, and the next question is to situate our analysis in the environment. We care about the ways in which the robot can use its control degrees of freedom to position itself in the environment. For instance, consider the Ackerman vehicle, or automobile. The total number of control degrees of freedom for such a vehicle is  $\delta_M = 2$ , one for steering and the second for actuation of the drive wheels. But what is the total degrees of freedom of the vehicle in its environment? In fact it is three: the car can position itself on the plane at any  $x, y$  point and with any angle  $\theta$ .

Thus identifying a robot's space of possible configurations is important because surprisingly it can exceed  $\delta_M$ . In addition to *workspace*, we care about how the robot is able to move between various configurations: what are the types of paths that it can follow and, furthermore, what are its possible trajectories through this configuration space? In the remainder of this discussion, we move away from inner kinematic details such as wheels and focus instead on the robot chassis pose and the chassis degrees of freedom. With this in mind, let us place the robot in the context of its workspace now.

#### 3.4.1 Degrees of freedom

In defining the workspace of a robot, it is useful to first examine its *admissible velocity space*. Given the kinematic constraints of the robot, its velocity space describes the independent components of robot motion that the robot can control. For example, the velocity space of a unicycle can be represented with two axes, one representing the instantaneous forward speed of the unicycle and the second representing the instantaneous change in orientation,  $\dot{\theta}$ , of the unicycle.

The number of dimensions in the velocity space of a robot is the number of independently achievable velocities. This is also called the *differentiable degrees of freedom* (*DDOF*). A robot's *DDOF* is *always* equal to its degree of mobility  $\delta_m$ . For example, a bicycle has the following degree of maneuverability:  $\delta_M = \delta_m + \delta_s = 1 + 1 = 2$ . The *DDOF* of a bicycle is indeed 1.